

# MTH102

## Linear Algebra

2025–2026–I SEMESTER

### PROBLEM SETS

18 September 2025

#### Chapter 1. Problem set 1

**Exercise 1.1.** Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$ .
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$ .

**Exercise 1.2.** What does it mean to say that  $A$  is not a subset of  $B$ ?

**Exercise 1.3.** Show that if  $A$  is a subset of  $B$  and  $B$  is a subset of  $C$ , then  $A$  is a subset of  $C$ .

**Exercise 1.4.** Show that no element  $n$  of  $\mathbb{N}$  satisfies  $n^4 - 5n^2 + 6 = 0$ .

**Exercise 1.5.** What does it mean to say that two sets  $A, B$  are not equal?

**Exercise 1.6.** Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

**Exercise 1.7.** Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

**Exercise 1.8.** If  $A, B$  are sets, show that

$$A \subseteq A \cup B, B \subseteq A \cup B.$$

**Exercise 1.9.** If  $A, B$  are subsets of a set  $C$ , then  $A \cup B$  is a subset of  $C$ .

**Exercise 1.10.** If  $A$  is a set, show that

$$A \cup A = A.$$

**Exercise 1.11** (Commutative property). If  $A, B$  are sets, show that

$$A \cup B = B \cup A.$$

**Exercise 1.12** (Associative property). If  $A, B, C$  are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

**Exercise 1.13.** Determine the intersection of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

**Exercise 1.14.** If  $A, B$  are sets, show that

$$A \cap B \subseteq A, A \cap B \subseteq B.$$

**Exercise 1.15.** If  $A, B, C$  are sets satisfying

$$C \subseteq A, C \subseteq B,$$

then show that

$$C \subseteq A \cap B$$

holds.

**Exercise 1.16.** Identify the integers among

$$1, 2, 3, 4, \dots, 20$$

which are divisible by at least one of 2 and 5.

**Exercise 1.17** (Inclusion-exclusion principle). Show that for finite subsets  $A, B, C$  of a set  $X$ ,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

**Exercise 1.18.** Determine the number of integers among

$$1, 2, 3, 4, \dots, 100,$$

which are divisible by at least one of 6, 10, 15.

**Exercise 1.19.** If  $A$  is a set, show that

$$A \cap A = A.$$

**Exercise 1.20** (Commutative property). If  $A, B$  are sets, show that

$$A \cap B = B \cap A.$$

**Exercise 1.21** (Associative property). If  $A, B, C$  are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

**Exercise 1.22** (Distributive property). If  $A, B, C$  are sets, show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

**Exercise 1.23.** Show that

$$A^c = \{x \in X : x \notin A\},$$

where  $X$  denotes the underlying universal set.

**Exercise 1.24.** Determine the complement of  $\{1, 2, 3, 4\}$  in  $\{1, 3, 5, 7\}$ , and the complement of  $\{1, 3, 5, 7\}$  in  $\{1, 2, 3, 4\}$ .

**Exercise 1.25.** If  $A, B$  are subsets of a set  $X$ , show that

$$A^c \cap B = B \setminus A.$$

**Exercise 1.26.** If  $A$  is a subset of a set  $X$ , then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

**Exercise 1.27.** If  $A, B$  are subsets of a set  $X$ , then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

**Exercise 1.28.** Show that for any sets  $A, B$ , the following are equivalent.

$$\begin{aligned} A &\subseteq B, \\ A \cap B &= A, \\ A \cup B &= B, \\ B^c &\subseteq A^c. \end{aligned}$$

**Exercise 1.29.** If  $A, B, C$  are sets, show that

1.  $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C),$
2.  $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$

**Exercise 1.30.** Let  $A, B$  be subsets of a set  $X$ . Show that the sets

$$A \setminus B, B \setminus A$$

are disjoint.

**Exercise 1.31.** Determine the symmetric difference of  $\{1, 2, 3, 4\}$  and  $\{1, 3, 5, 7\}$ .

**Exercise 1.32.** If  $A, B$  are sets, then show that

$$\begin{aligned} A \Delta B &= B \Delta A, \\ A \cup B &= (A \Delta B) \cup (A \cap B), \end{aligned}$$

and

$$(A \Delta B) \cap (A \cap B) = \emptyset.$$

**Exercise 1.33.** Do there exist three finite sets  $A, B, C$  satisfying

$$|A \Delta B| = 1, |B \Delta C| = 2, |C \Delta A| = 4?$$

**Exercise 1.34.** If  $A, B$  are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set  $X$ , its number of elements is denoted by  $|X|$ .

**Exercise 1.35.** Determine the cartesian product of  $[1, 2]$  and  $[3, 4] \cup [5, 6]$ .

**Exercise 1.36.** Identify the set

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

**Exercise 1.37.** Let  $A, B$  be subsets of  $\mathbb{R}$  defined by

$$A = \{x \in \mathbb{R} : x^2 \geq 0\},$$

$$B = \{x \in \mathbb{R} : x^3 \geq 0\}.$$

Determine the sets  $A \cup B, A \cap B, A \setminus B, B \setminus A$ .

**Exercise 1.38.** For a positive integer  $k$ , let  $A_k$  denote the set of integral multiples of  $k$ , that is,

$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$

Let  $m, n$  be positive integers. For each of the following statements, determine the equivalent conditions on the integers  $m, n$ .

1.  $A_m \subseteq A_n$
2.  $A_m \subsetneq A_n$
3.  $A_m \not\subseteq A_n$
4.  $A_m = A_n$
5.  $A_m \cap A_n = \emptyset$
6.  $A_m \setminus A_n \neq \emptyset$
7.  $A_m \setminus A_n \neq \emptyset$  or  $A_n \setminus A_m \neq \emptyset$
8.  $A_m \setminus A_n \neq \emptyset$  and  $A_n \setminus A_m \neq \emptyset$

**Exercise 1.39.**

## Chapter 2. Induction principles

**Exercise 2.1.** If  $A$  is a nonempty subset of  $\mathbb{N}$ , and  $a_1, a_2$  are least elements of  $A$ , then show that  $a_1 = a_2$ .

**Exercise 2.2** (Infinitude of primes, by Saidak). Let  $a_1 = 2$ , and  $a_{n+1} = a_n(a_n + 1)$  for any positive integer  $n$ . For any  $n \in \mathbb{N}$ , show that  $a_n$  has at least  $n$  distinct prime divisors.

**Exercise 2.3.** Show that there are infinitely many prime numbers of the form  $4k + 3$ , where  $k$  is a nonnegative integer.

**Exercise 2.4.** Show that

$$2^{2^n} - 1$$

has at least  $n$  distinct prime divisors for all  $n \in \mathbb{N}$ .

**Exercise 2.5.** Show that every positive integer greater than 1 is a prime or is a product of prime numbers.

**Exercise 2.6.** Let  $S$  be a finite set of size  $n$ . Show that the number of subsets of  $S$  is  $2^n$ .

**Exercise 2.7.** Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all  $n \in \mathbb{N}$ .

**Exercise 2.8.** Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all  $n \in \mathbb{N}$ .

**Exercise 2.9.** Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all  $n \in \mathbb{N}$ .

**Exercise 2.10.** Show that

$$3 + 11 + \cdots + (8n-5) = 4n^2 - n$$

for all  $n \in \mathbb{N}$ .

**Exercise 2.11.** Show that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$$

for all  $n \in \mathbb{N}$ .

**Exercise 2.12.** Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all  $n \in \mathbb{N}$ .

**Exercise 2.13.** Show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left( \frac{n(n+1)}{2} \right)^2$$

hold for any positive integer  $n$ .

**Exercise 2.14.** Show that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

for all  $n \in \mathbb{N}$ .

**Exercise 2.15.** Show that  $n^3 + 5n$  is divisible by 6 for all  $n \in \mathbb{N}$ .

**Exercise 2.16.** Show that  $5^{2n} - 1$  is divisible by 8 for all  $n \in \mathbb{N}$ .

**Exercise 2.17.** Show that  $5^n - 4n - 1$  is divisible by 16 for all  $n \in \mathbb{N}$ .

**Exercise 2.18.** Show that  $6^n - 5n - 1$  is divisible by 25 for all  $n \in \mathbb{N}$ .

**Exercise 2.19.** Show that  $n^3 + (n+1)^3 + (n+2)^3$  is divisible by 9 for all  $n \in \mathbb{N}$ .

**Exercise 2.20.** Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all  $n \in \mathbb{N}$  satisfying  $n > 1$ .

**Exercise 2.21.** Show that  $3^n \geq n^2$  for all  $n \in \mathbb{N}$ .

**Exercise 2.22.** Show that  $n! > 2^n$  for all  $n \in \mathbb{N}$  satisfying  $n \geq 4$ .

**Exercise 2.23.** Show that

$$\left(1 + \frac{1}{1^3}\right) \left(1 + \frac{1}{2^3}\right) \left(1 + \frac{1}{3^3}\right) \cdots \left(1 + \frac{1}{n^3}\right) < 3$$

holds for all  $n \in \mathbb{N}$ .

**i Remark**

Does taking  $P(n)$  to be the statement that

$$\left(1 + \frac{1}{1^3}\right)\left(1 + \frac{1}{2^3}\right)\left(1 + \frac{1}{3^3}\right)\cdots\left(1 + \frac{1}{n^3}\right) < 3$$

help?

**Exercise 2.24.** Show that

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3} < \frac{3}{2}$$

for all  $n \in \mathbb{N}$ .

**Exercise 2.25.** Show that

$$\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{2^2}\right)\left(1 + \frac{1}{2^3}\right)\cdots\left(1 + \frac{1}{2^n}\right) < \frac{5}{2}$$

holds for all  $n \in \mathbb{N}$ .

Here are a few exercises where strong induction is useful.

**Exercise 2.26.** Let the integers  $x_1, x_2, \dots$  be defined by

$$x_1 := 1,$$

$$x_2 := 2,$$

$$x_{n+2} := \frac{1}{2}(x_n + x_{n+1})$$

for all  $n \in \mathbb{N}$ . Show that  $1 \leq x_n \leq 2$  for all  $n \in \mathbb{N}$ .

**Exercise 2.27.** The Fibonacci sequence  $F_0, F_1, F_2, \dots$  is defined by

$$F_0 := 0,$$

$$F_1 := 1,$$

$$F_2 := 1,$$

$$F_{n+2} := F_n + F_{n+1}$$

for all  $n \in \mathbb{N}$ . Show that

$$F_n = \frac{1}{\sqrt{5}} \left( \left( \frac{1 + \sqrt{5}}{2} \right)^n - \left( \frac{1 - \sqrt{5}}{2} \right)^n \right)$$

for all  $n \in \mathbb{N} \cup \{0\}$ .

**Exercise 2.28.** Show that

$$F_0^2 + F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$$

for all  $n \in \mathbb{N} \cup \{0\}$ .

## Chapter 3. Matrices

**Exercise 3.1.** If  $A, B, C$  are matrices of the same size, show that

$$A + (B + C) = (A + B) + C.$$

**Exercise 3.2.** If  $A, B$  are matrices of the same size, show that

$$A + B = B + A.$$

**Exercise 3.3.** If  $A$  is an  $m \times n$  matrix, show that

$$A + 0_{m \times n} = 0_{m \times n} + A = A$$

holds, where  $0_{m \times n}$  denotes the  $m \times n$  zero matrix, that is, the  $m \times n$  matrix whose entries are all zero.

**Exercise 3.4.** Show that if  $A$  is a matrix and  $c, d$  are real numbers, then

$$(c + d)A = cA + dA$$

and

$$c(dA) = (cd)A.$$

**Exercise 3.5.** If  $A$  is a matrix, what is  $(A^T)^T$ ?

**Exercise 3.6.** Show that if  $A, B, C$  are matrices such that the products  $A(BC)$  and  $(AB)C$  are defined, then

$$A(BC) = (AB)C.$$

**Exercise 3.7.** Provide examples to show that in general, matrix multiplication is not commutative, that is,  $AB \neq BA$  for some matrices  $A, B$ .

**Exercise 3.8 (\*\*).** Show that for any  $n \in \mathbb{N}$  with  $n \geq 2$ , there are  $n \times n$  matrices  $A, B$  such that

$$AB \neq BA.$$

### ? Question

Can induction be used for the above exercise?

**Exercise 3.9.** If  $A$  is an  $n \times n$  matrix, show that

$$AI_n = I_n A = A$$

holds, where  $I_n$  denotes the  $n \times n$  diagonal matrix, with all diagonal entries equal to 1, that is,

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

**Exercise 3.10.** If  $A$  is an  $m \times n$  matrix, show that

$$I_m A = A, A I_n = A$$



where  $I_m$  (resp.  $I_n$ ) denote the  $m \times m$  (resp.  $n \times n$ ) identity matrix.

**Exercise 3.11.** If  $A$  is an  $m \times n$  matrix with entries in  $\mathbb{R}$  and  $c$  is a real number, then

$$cA = (cI_n)A = A(cI_m),$$

where  $I_n$  (resp.  $I_m$ ) denotes the  $n \times n$  (resp.  $m \times m$ ) identity matrix.

**Exercise 3.12.** Compute

$$\begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}^2.$$

**Exercise 3.13.** Show that if  $A, B$  are matrices such that the products  $AB$  and  $B^T A^T$  are defined, then

$$(AB)^T = B^T A^T.$$

**Exercise 3.14.** Show that if  $A, B, C$  are matrices such that the sum  $B + C$ , and the products  $AB, AC, A(B + C)$  are defined, then

$$A(B + C) = AB + AC.$$

**Exercise 3.15.** Show that if  $A, B, C$  are matrices such that the sum  $B + C$ , and the products  $AB, AC, A(B + C)$  are defined, then

$$(A + B)C = AC + BC.$$

**Exercise 3.16.** If  $A$  is an  $m \times n$  matrix, then show that for all  $i = 1, 2, \dots, n$ , the product  $Ae_i$  is equal to the  $i$ -th column of  $A$ , where  $e_i$  denotes the  $i$ -th standard basis vector of  $\mathbb{R}^n$ , that is,

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

with 1 in the  $i$ -th position, and 0's elsewhere.

Let us revisit [Exercise 3.12](#), under the light of [Exercise 3.16](#).

**Exercise 3.17.** Compute  $B^2$  where  $B$  is the following  $n \times n$  matrix

$$B = \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}.$$

**Exercise 3.18.** Show that if  $B, C$  are inverses of an  $n \times n$  matrix  $A$ , then  $B = C$ .

**Exercise 3.19.** Let  $A, B$  be invertible  $n \times n$  matrices. Show that  $AB$  is invertible, and that

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Compare the above exercise with [Exercise 3.13](#).

**Exercise 3.20.** If  $A, B$  are  $2 \times 2$  matrices, then show that

$$\det(AB) = \det(A) \det(B).$$

**Exercise 3.21.** Solve the system of equations

$$5x + 2y = 11,$$

$$3x + 4y = 8.$$

**Exercise 3.22.** Solve the system of equations

$$x + 2y + 3z = 1,$$

$$4x + 5y + 6z = 2,$$

$$7x + 8y + 9z = 3.$$

**Exercise 3.23.** Let  $A, B$  be  $2 \times 2$  matrices. Show that if  $AB$  is invertible, then both  $A$  and  $B$  are invertible.

**Exercise 3.24.** Let  $A$  be an  $n \times n$  matrix. If  $P$  is an invertible  $n \times n$  matrix, then show that

$$(PAP^{-1})^k = PA^kP^{-1}$$

holds for any positive integer  $k$ .